

Pseudocode Algoritma NMPC

Algoritma 1: Simulasi NMPC Konvensional Kapal Otonom

Masukan: Parameter kapal $(L, B, T, m, u_0, C_B, x_G, \rho)$, parameter NMPC $(T_p, T_{\text{sim}}, T_{\text{sim,total}}, \mathbf{Q}, \mathbf{R})$, batasan fisik $(r_{\text{limit}}, u_{\text{limit}}, u_{\text{rate,limit}})$, kondisi awal $\mathbf{s}_{0,\text{dim}}$, dan rute referensi $\mathbf{h}_{\text{ref}}$.

Keluaran: Riwayat state $\mathbf{S}_{\text{dim}}$, riwayat input $\mathbf{U}_{\text{applied}}$, total waktu komputasi t_{total} , dan error RMSE (e_X, e_Y, e_Ψ) .

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1  $t \leftarrow 0$ ;
2  $\mathbf{s}(0) \leftarrow \mathbf{s}_{0,\text{nd}}$ ;
3  $\text{history\_state} \leftarrow [\mathbf{s}(0)]$ ,  $\text{history\_input} \leftarrow []$ ,  $u_{\text{prev}} \leftarrow 0$ ;
4 Hitung  $N \leftarrow \text{round}(T_p/T_{\text{sim}})$  dan koefisien hidrodinamika  $Y_{\dot{v}}, Y_{\dot{r}}, N_{\dot{v}}, N_{\dot{r}}, Y_v, Y_r, N_v, N_r$ ;
5 Hitung parameter non-dimensional  $m_{\text{nd}} \leftarrow \frac{2m}{\rho L^3}$ ,  $x_{G,\text{nd}} \leftarrow \frac{x_G}{L}$ ,  $I_{z,\text{nd}} \leftarrow 1.2392 \times 10^{-4}$ ,  $u_{0,\text{nd}} \leftarrow 1$ ;
6 Bentuk matriks massa  $M_{\text{mat}}$  serta matriks sistem  $A_{\text{sys}}$  dan  $B_{\text{sys}}$ ;
7 Hitung batasan non-dimensional  $r_{\text{limit,nd}} \leftarrow r_{\text{limit}} \cdot (L/u_0)$  dan  $du_{\text{max}} \leftarrow u_{\text{rate,limit}} \cdot T_{\text{sim}}$ ;
8 Bentuk matriks kendala linear laju kemudi  $[A_{\text{du}}, b_{\text{du}}] \leftarrow \text{du\_constraints}(N, u_{\text{prev}}, du_{\text{max}})$ ;
9 Tentukan batas atas dan bawah input  $\mathbf{lb} \leftarrow -u_{\text{limit}} \cdot \mathbf{1}_N$  dan  $\mathbf{ub} \leftarrow u_{\text{limit}} \cdot \mathbf{1}_N$ ;
10 Atur konfigurasi optimizer SQP options  $\leftarrow \text{optimoptions('fmincon', 'Algorithm', 'sqp', ...)}$ ;
11 while  $t \leq T_{\text{sim,total}} - T_{\text{sim}}$  do
12     Dapatkan sekuen waktu prediksi  $t_{\text{pred}} \leftarrow t + [1, 2, \dots, N] \cdot T_{\text{sim}}$ ;
13     Hitung sekuen lintasan referensi  $x_{\text{ref,seq}} \leftarrow (h_{\text{ref}}(1) + t_{\text{pred}} \cdot u_0)/L$ ,
         $y_{\text{ref,seq}} \leftarrow (h_{\text{ref}}(2)/L) \cdot \mathbf{1}_N$ ,  $\psi_{\text{ref,seq}} \leftarrow h_{\text{ref}}(3) \cdot \mathbf{1}_N$ ;
14     Formulasi fungsi biaya  $J(U) = \text{mpc\_cost}(U, \mathbf{s}_{\text{nd}}, u_{\text{prev}}, x_{\text{ref,seq}}, y_{\text{ref,seq}}, \psi_{\text{ref,seq}}, \dots)$ ;
15     Formulasi kendala nonlinear  $c(U) = \text{state\_constraints}(U, \mathbf{s}_{\text{nd}}, r_{\text{limit,nd}}, \dots)$ ;
16     Set tebakan awal sekuen kontrol  $U_0 \leftarrow u_{\text{prev}} \cdot \mathbf{1}_N$ ;
17     Selesaikan optimasi SQP
         $[U_{\text{opt}}, \sim, \text{exitflag}] \leftarrow \text{fmincon}(J, U_0, A_{\text{du}}, b_{\text{du}}, [], [], \mathbf{lb}, \mathbf{ub}, c, \text{options})$ ;
18     if  $\text{exitflag} \leq 0$  then
19         Tetapkan solusi cadangan  $U_{\text{opt}} \leftarrow U_0$ ;
20     Ambil input langkah pertama  $u_{\text{applied}} \leftarrow U_{\text{opt}}(1)$ ;
21     Simpan variabel input  $\text{history\_input} \leftarrow [\text{history\_input}, u_{\text{applied}}]$ ;
22     Propagasi keadaan kapal  $\mathbf{s}_{\text{nd}} \leftarrow \text{euler\_step}(\text{ship\_dynamics}, \mathbf{s}_{\text{nd}}, u_{\text{applied}}, dt_{\text{nd}})$ ;
23     Simpan variabel keadaan  $\text{history\_state} \leftarrow [\text{history\_state}, \mathbf{s}_{\text{nd}}]$ ;
24     Update  $u_{\text{prev}} \leftarrow u_{\text{applied}}$  dan hitung ulang  $[A_{\text{du}}, b_{\text{du}}] \leftarrow \text{du\_constraints}(N, u_{\text{prev}}, du_{\text{max}})$ ;
25      $t \leftarrow t + T_{\text{sim}}$ ;
26 Penyelarasan riwayat input  $\text{history\_input} \leftarrow [\text{history\_input}, \text{history\_input}(\text{end})]$ ;
27 Konversi riwayat keadaan ke dimensional
     $\mathbf{S}_{\text{dim}} \leftarrow \text{nondimensional\_to\_dimensional}(\text{history\_state}, L, u_0)$ ;
28 Hitung error RMSE  $(e_X, e_Y, e_\Psi)$  terhadap lintasan referensi penuh;
29 Plot grafik lintasan kapal, yaw rate, sudut kemudi  $u$ , dan laju perubahan kemudi  $\Delta u$ ;

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Algoritma 2: Fungsi Pendukung NMPC

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1 Fungsi euler_step( $f, x, u, dt$ ):
2   return  $x + dt \cdot f(x, u)$ ;

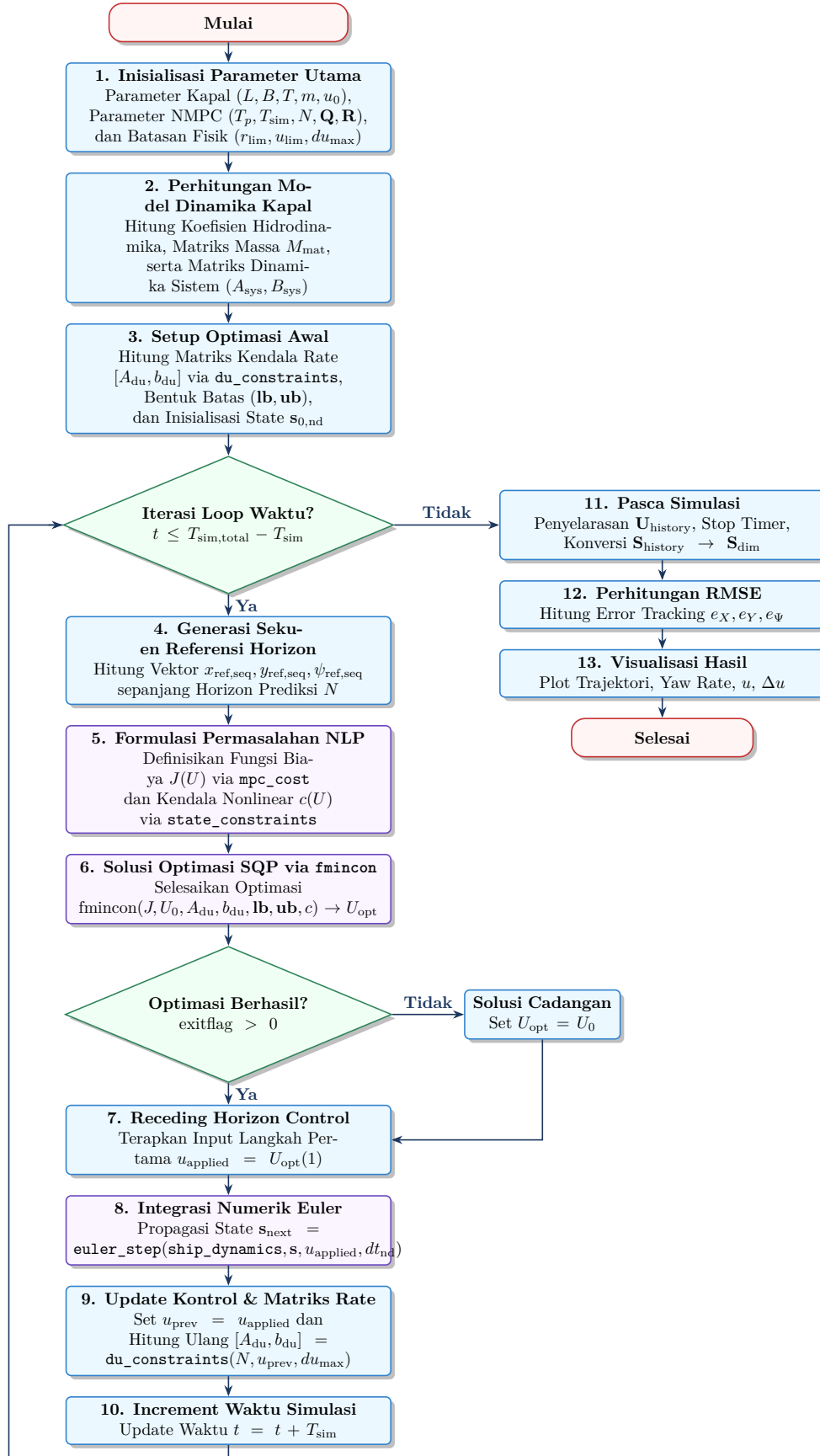
3 Fungsi ship_dynamics( $s, u, A_{sys}, B_{sys}, u_{0,nd}$ ):
4   Ekstrak state:  $v \leftarrow s(1), r \leftarrow s(2), \psi \leftarrow s(5)$ ;
5   Dinamika sway-yaw:  $[\dot{v}; \dot{r}] \leftarrow A_{sys}[v; r] + B_{sys}u$ ;
6   Kinematika posisi:  $\dot{x} \leftarrow u_{0,nd} \cos(\psi) - v \sin(\psi), \quad \dot{y} \leftarrow u_{0,nd} \sin(\psi) + v \cos(\psi)$ ;
7   return  $[\dot{v}; \dot{r}; \dot{x}; \dot{y}; r]$ ;

8 Fungsi mpc_cost( $U, s_0, u_{prev}, x_{ref,seq}, y_{ref,seq}, \psi_{ref,seq}, T_{sim}, L, u_0, A_{sys}, B_{sys}, u_{0,nd}, Q, R$ ):
9    $N \leftarrow \text{length}(U), \quad s \leftarrow s_0, \quad J \leftarrow 0, \quad dt_{nd} \leftarrow (T_{sim} \cdot u_0)/L$ ;
10  for  $i \leftarrow 1$  to  $N$  do
11     $u \leftarrow U(i)$ ;
12     $s \leftarrow \text{euler\_step}(\text{ship\_dynamics}, s, u, dt_{nd})$ ;
13     $err \leftarrow [s(3) - x_{ref,seq}(i); s(4) - y_{ref,seq}(i); s(5) - \psi_{ref,seq}(i)]$ ;
14     $J \leftarrow J + err^T Q err + R \cdot u^2$ ;
15  return  $J$ ;

16 Fungsi state_constraints( $U, s_0, T_{sim}, L, u_0, A_{sys}, B_{sys}, u_{0,nd}, r_{limit,nd}$ ):
17   $N \leftarrow \text{length}(U), \quad s \leftarrow s_0, \quad c \leftarrow [], \quad dt_{nd} \leftarrow (T_{sim} \cdot u_0)/L$ ;
18  for  $i \leftarrow 1$  to  $N$  do
19     $s \leftarrow \text{euler\_step}(\text{ship\_dynamics}, s, U(i), dt_{nd})$ ;
20     $r \leftarrow s(2)$ ;
21     $c \leftarrow [c; r - r_{limit,nd}; -r_{limit,nd} - r]$ ;
22  return  $(c, [])$ ; // Tidak ada kendala kesamaan ( $ceq = []$ )

23 Fungsi du_constraints( $N, u_{prev}, du_{max}$ ):
24  Inisialisasi matriks  $A \leftarrow \mathbf{0}_{2N \times N}$  dan vektor  $b \leftarrow \mathbf{0}_{2N \times 1}$ ;
25   $A(1,1) \leftarrow 1, \quad b(1) \leftarrow u_{prev} + du_{max}$ ;
26  for  $i \leftarrow 2$  to  $N$  do
27     $A(i, i-1) \leftarrow -1, \quad A(i, i) \leftarrow 1, \quad b(i) \leftarrow du_{max}$ ;
28   $A(N+1,1) \leftarrow -1, \quad b(N+1) \leftarrow du_{max} - u_{prev}$ ;
29  for  $i \leftarrow 2$  to  $N$  do
30     $A(N+i, i-1) \leftarrow 1, \quad A(N+i, i) \leftarrow -1, \quad b(N+i) \leftarrow du_{max}$ ;
31  return  $(A, b)$ ;
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Diagram Alir NMPC



Gambar 1: Diagram alir NMPC